

ORIGINAL RESEARCH

Navigating the methodological, ethical, and operational challenges of trials within cohorts: insights from a French pediatric research-based cohort

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Abstract

Background: Trials within Cohorts (TwICs) provide a pragmatic framework to improve recruitment and external validity by embedding randomized controlled trials within longitudinal observational cohorts. In pediatrics, and particularly in rare diseases, this design is especially attractive but remains difficult to implement in routine practice due to intertwined methodological, ethical, and operational complexities. Building on the development of a national pediatric platform for idiopathic nephrotic syndrome in France, this commentary highlights key challenges and proposes practical strategies to address them.

Key challenges: We focus on three domains: (i) informed consent, with particular attention to the ethical and regulatory challenges raised by staged information and deferred consent approaches; (ii) data quality, including the long-term requirements for high-quality, interoperable, and research-ready clinical data; and (iii) sustainability, covering governance and funding models needed to maintain a cohort-based trial infrastructure beyond short-term, project-driven grants.

Practical strategies: We argue that scalable pediatric TwICs would benefit from secure and family-centered e-consent solutions, robust real-world data infrastructures aligned with interoperability standards, and hybrid funding strategies.

Conclusions: Overall, this commentary offers practical strategies to help teams overcome key ethical, methodological, and operational challenges to implementing pediatric TwICs. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Trials within cohorts (TwICs); Pediatrics; Pragmatic trials; Informed consent; Real-world evidence; Research infrastructure

1. Introduction

Randomized controlled trials (RCTs) are the gold standard for evaluating the efficacy and safety of medical interventions [1]. Their methodological rigor, grounded in

random allocation and controlled comparison, provides strong internal validity and supports causal inference. However, conventional stand-alone RCTs may face limitations that reduce their applicability to routine clinical practice. Recruitment difficulties, restrictive eligibility criteria, substantial drop-out, and burdensome procedures (additional study visits, frequent assessments, and monitoring requirements) often lead to highly selected study populations, thereby limiting external validity and generalizability [2,3].

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Plain Language Summary

Trials within Cohorts (TwICs) represent an innovative approach to clinical research where randomized trials are embedded directly into existing groups of patients, known as cohorts. By using this framework, researchers can ensure that their findings more accurately reflect real-world medical practice. This design is particularly valuable in fields like pediatrics and the study of rare diseases, where the limited number of available patients often makes traditional trial recruitment exceptionally difficult.

Despite their potential, implementing TwICs involves practical complexities that require careful management. One primary challenge involves the information and consent process, as researchers must find ways to properly inform patients and families while complying with strict national ethical and regulatory frameworks. In addition, ensuring the accuracy and seamless sharing of long-term medical data remains a significant technical struggle, as routine clinical data require substantial curation to be "research-ready." There is also the persistent issue of financial sustainability, as traditional funding models often prioritize individual short-term projects rather than the long-term maintenance required to keep a research infrastructure functional over many years.

Drawing from experience with a national platform for children with kidney disease in France, several strategic solutions can help overcome these difficulties. Utilizing digital tools such as electronic consent can streamline the enrollment process, improve the delivery of information, and make it more accessible for families to participate. Furthermore, investing in robust and interoperable data systems ensures that high-quality information is consistently available for analysis. Finally, a shift in funding strategy is essential, moving away from temporary grants toward sustained support that recognizes cohorts as vital research infrastructures. By adopting these improvements, TwICs can become a powerful and efficient tool for evaluating how medical treatments perform in everyday clinical settings.

To address these challenges, the trials within cohorts (TwICs) design has emerged as an innovative approach that embeds randomized trials within large, well-characterized prospective observational cohorts [2]. By embedding trials in existing cohort infrastructure for participant identification, follow-up, and outcome collection, TwICs improves representativeness, streamlines recruitment, and reduces operational complexity.

Although TwICs has been implemented in several countries, including the United Kingdom, the Netherlands, Canada, and Australia [4–10], it remains uncommon in France outside the exceptional context of the COVID-19 pandemic [8,11,12]. Drawing on our experience developing a national pediatric TwICs platform for idiopathic nephrotic syndrome [13], this article examines the main challenges to implementing TwICs in a national context and discusses pragmatic approaches that may facilitate their use, focusing on informed consent, data quality, and sustainability.

2. The trials within cohorts model: principles, promises, and limitations

This model relies on a large prospective observational cohort characterized at baseline and followed longitudinally, with systematic collection of health outcomes and relevant clinical parameters. Rather than being structured around a single intervention, the cohort constitutes a long-term research infrastructure intended to support multiple embedded trials over time.

Within the cohort, eligibility criteria are defined for each nested trial. Among eligible participants, a subset is

randomly selected and offered the intervention under evaluation, while the remaining participants continue to receive usual care and serve as controls. This design ensures that intervention and control groups originate from the same real-world population and are followed within the same health care context, thereby enhancing external validity and comparability.

A defining and frequently debated feature of the original TwICs model proposed by Relton et al. [2] is the staging of consent. At cohort enrollment, participants provide broad consent for cohort follow-up and for the possibility of being randomly selected for future embedded trials. Importantly, staged-consent models include explicit agreement that, if not selected for an intervention (ie, allocated to usual-care), they will not be contacted at trial launch. In the second stage, only patients randomly selected to the intervention arm receive trial-specific information and are asked for trial-specific consent. Therefore, patients agree upfront to the possibility that they may participate as controls in a trial without receiving additional notification. This “just-in-time” strategy offers a pragmatic alternative to traditional models by concentrating recruitment efforts, such as detailed information, documentation, and tracking, exclusively on the intervention arm, thereby enhancing operational efficiency [14].

Although staged consent characterizes the original TwICs proposal, informed consent procedures vary across trials conducted within cohorts [8]. Some use staged consent, whereas others apply conventional prerandomization consent while retaining the same core structural features: recruitment from a pre-established cohort, systematic outcome collection, and the possibility of multiple

What is new?**Key findings**

- Pediatric TwiCs must balance consent models, data quality, and sustainability.
- Cohorts are strategic, long-term research infrastructures.

What this adds to what was known?

- This study moves TwiCs beyond theory toward practical national implementation.
- It highlights dual-parent consent and evolving child decisional capacity.

What is the implication and what should change now?

- TwiCs platforms need opt-in e-consent models and dedicated structural funding.

embedded randomized comparisons. Importantly, this embedded structure differs from trials that use cohorts only to identify eligible patients without integrating randomization and follow-up within the cohort platform.

In this article, we use the term TwiCs in a broad structural sense to describe trials embedded within an established cohort infrastructure. When specifically referring to the staged-consent model proposed by Relton et al., this is explicitly stated.

Beyond consent-related considerations, TwiCs offer several methodological and practical advantages. By leveraging existing cohorts and routinely collected data, it may substantially reduce costs and timelines associated with participant recruitment, baseline assessments, and follow-up [6,15]. This is particularly relevant in pediatrics and in chronic or rare diseases, where stand-alone recruitment is often challenging. In addition, cohort-based enrollment tends to be broader and less selective than trial-specific recruitment, resulting in study populations that more closely reflect real-world clinical heterogeneity.

Despite these strengths, TwiCs also present limitations. A major methodological challenge arises when participants randomized to the intervention arm decline the proposed intervention [15–17]. In this design, randomization determines who is offered an intervention, not who ultimately receives it. Patients who refuse the intervention revert to usual care, which may dilute the contrast between study arms and reduce statistical power. If refusal is associated with patient characteristics, disease severity, or treatment preferences, selection bias may be introduced within the intervention arm, complicating interpretation. In conventional stand-alone RCTs,

prerandomization consent typically yields more balanced nonadherence across arms, simplifying analysis. Consequently, TwiCs should be viewed as complementary to, rather than a replacement for, conventional RCTs. It appears particularly well suited for pragmatic evaluations in real-world settings, whereas traditional RCTs remain essential for studies requiring strict internal validity, such as pivotal confirmatory phase III trials, particularly when placebo is used as a comparator.

3. Challenges to practical implementation: example of the French national context

Despite the compelling methodological advantages of the TwiCs design, its widespread adoption can face significant hurdles. These challenges stem from national regulatory frameworks, fragmented health data infrastructures, and entrenched financial and organizational constraints. Addressing these issues is paramount to fully realizing TwiCs' potential.

3.1. Information and consent

A primary challenge to TwiCs implementation stems from robust national legal and ethical frameworks surrounding informed consent. The deferred consent approach that characterizes the original TwiCs design conflicts with the principles of Good Clinical Practice, which require informed consent to be obtained prior to participation in research and that participants are provided with clear and comprehensive information about the study, including its objectives, randomization procedures, and potential risks and benefits [18]. In a TwiCs framework, cohort participants allocated to the control group are not informed about the specific randomized intervention for which their data are used as a comparator.

This approach raises ethical concerns regarding respect for patient autonomy, a core principle of medical research articulated in the Declaration of Helsinki [19]. Autonomy implies not only the freedom to decide whether to participate in research, but also access to sufficient information to make that decision meaningfully. It has been argued that withholding trial-specific information from control participants may undermine this principle, even when data collection is limited to routine care and does not impose additional burden [20].

This issue is further highlighted by studies investigating patient recall of broad consent in TwiCs. For instance, Verweij et al. conducted a study evaluating the experience of patients who served as controls in TwiCs designs without notification at the time of randomization. Two to 6 years after cohort entry, only 15% of all patients remembered having given broad consent to randomization [21]. This result is problematic, as comprehensive and truly informed consent is an essential condition for the legal and ethical

conduct of any research involving human subjects. The low recall rate raises serious questions about the continuous validity of consent over time and about whether patients fully grasp the implications of their initial broad consent, especially concerning their potential inclusion in future, undisclosed randomized trials. Therefore, French regulations strongly prioritize explicit, trial-specific, and contemporaneous consent.

These issues are amplified in pediatric research: if adult participants often fail to recall what they consented to years earlier [21], the persistence of parental consent given on behalf of a child is questionable, particularly as the child reaches adolescence and gains decisional capacity. In addition, the accumulation of consent procedures for successive embedded trials creates a significant operational challenge. Although both parents are generally expected to provide written consent [22], it frequently happens that only one parent attends the consultation where the investigating physician presents the study and addresses the questions of both the child and the parent who is present. Providing information to the second parent requires organizing a separate information session, which can be challenging for professionals already overwhelmed by their daily activities.

In pediatric research, authorization from both parents is generally required. According to French regulation, single-parent authorization is permitted only under strict minimal-risk conditions and when the second parent's authorization cannot be obtained in time [23]. These criteria are rarely met in cohorts as inclusion is usually not time-critical, and medicinal product trials generally exceed the minimal-risk threshold.

While this requirement applies to pediatric research in general and is not specific to the TwiCs design, the combination of this dual parental authorization necessity and the regulatory preference for trial-specific consent in interventional studies discussed above makes the original staged-consent model particularly challenging to implement in the French pediatric setting.

3.2. Data quality and completeness

Ensuring longitudinal data quality and completeness is a major challenge for TwiCs, which require systematic, research-grade outcome collection with standardized definitions and schedules. A key distinction must be made between routinely collected care or medico-administrative data and prospectively collected cohort data designed for research. While routine data can support follow-up and outcome ascertainment, it is rarely sufficient for research without substantial curation, validation, and upfront standardization of variables and procedures [24–26]. Poorly specified data elements may require retrospective collection, increasing costs and potentially generating “missing-not-at-random” data and biased estimates. These issues are amplified by the scale and duration of cohorts.

Long-term follow-up produces large data volumes that preclude exhaustive verification.

Pediatric cohorts raise additional challenges, including age-dependent outcomes, shifts from proxy to self-reporting, evolving child assent requirements, and the need to harmonize measures across developmental stages and transitions to adult care.

In TwiCs, trial inclusion and follow-up must strictly align with existing cohort assessment schedules, creating a ‘synchronization constraint’ that often forces a trade-off between recruitment velocity and data freshness [15]. Unlike conventional stand-alone RCTs where participants are approached for consent and specific baseline measures can be collected at the time of randomization, the TwiCs approach typically relies on precollected cohort data to maintain its pragmatic nature. Investigators must therefore either restrict recruitment to narrow windows immediately following cohort waves or risk using outdated baseline measures.

In pediatrics, where rapid physiological and developmental changes occur, this misalignment can introduce significant baseline imbalance [27]. While supplemental ad hoc assessments could theoretically mitigate this, they would diminish the economic advantages of the TwiCs design, highlighting a core tension between TwiCs efficiency and clinical precision.

The 2025 report from the French Agence de l’Innovation en Santé highlights the conditions required for cohorts and registries to support regulatory evaluations, including defined responsibilities for data management, quality control, monitoring, and systematic handling of missing or inconsistent data [28]. Achieving these standards requires substantial resources, yet cohorts are often funded as low-risk observational research.

3.3. Financial and organizational constraints

Beyond ethical and data-related challenges, the implementation and long-term sustainability of TwiCs platforms face significant financial and organizational constraints. Maintaining a cohort over many years entails recurrent costs related to patient follow-up, data collection, data management, governance, and technical infrastructure. Unlike stand-alone trials, TwiCs platforms require continuous investment.

Across Europe, medical research funding is characterized by fragmentation and a predominance of short-term project-based grants [29]. In pediatrics, these constraints are often amplified by small eligible populations and the need for multicenter coordination. While such mechanisms are well suited to discrete studies, they are poorly adapted to supporting long-term research infrastructures such as cohorts. This challenge is particularly acute for noncommercial research, which often lacks alternative funding sources [29].

In France, historically limited investment in noncommercial biomedical research further exacerbates these difficulties [30]. Current national funding schemes, including the Program Hospitalier de Recherche Clinique and calls from the Agence Nationale de la Recherche, are primarily structured around traditional stand-alone trials [31,32]. Their timelines, budget structures, and evaluation criteria make it difficult to integrate the cumulative and multitrial nature of TwiCs platforms. As a result, research teams may struggle to secure stable funding for cohort maintenance, threatening the continuity and quality of data collection.

4. Solutions and perspectives for trials within cohorts implementation

Despite these challenges, several pragmatic solutions may facilitate the broader adoption of TwiCs platforms.

4.1. Choosing the informed consent model: staged or conventional consent

The choice between staged consent and conventional prerandomization consent reflects distinct ethical and methodological positions. In contexts where the staged-consent model is ethically debated or difficult to reconcile with national regulatory requirements, investigators may opt to provide trial-specific information to all eligible participants prior to randomization, obtaining explicit consent irrespective of allocation.

Adopting conventional prerandomization consent modifies the ethical architecture of participant information but does not alter the underlying cohort-embedded structure of the trial. The cohort continues to function as a structured recruitment reservoir and longitudinal follow-up platform. This preserves key infrastructural advantages, including efficient identification of eligible participants, improved representativeness from a well-defined population base, reduced reliance on trial-specific recruitment pathways, and streamlined baseline and outcome data collection.

Methodologically, conventional prerandomization consent may reduce concerns about differential refusal after allocation and can simplify statistical planning, as nonadherence is more likely to be balanced across groups [15]. Informing both groups may also facilitate the implementation of additional trial-specific assessments when required, increasing analytical flexibility while remaining aligned with prevailing ethical and regulatory standards. However, this approach may mitigate some advantages of the TwiCs design, as informing all eligible participants and obtaining consent prior to randomization may slow recruitment when refusals occur before allocation.

Conversely, staged consent may be appropriate where regulations permit broad cohort-level consent and where interventions are low risk and close to usual care. In such settings, limiting trial-specific information to participants

offered the intervention preserves the efficiency gains that motivated the TwiCs design.

4.2. Implementing electronic consent

Electronic consent represents another promising approach, particularly in pediatric research. Parents are often required to sign multiple consent forms over the course of a child's participation in a cohort and its nested trials, yet both parents may not be present at the same consultation. Secure electronic consent tools may facilitate information delivery and signature collection, while ensuring traceability and compliance with regulatory requirements [33]. Although qualified electronic signatures are legally equivalent to handwritten signatures, implementation is often complex and costly. In practice, advanced electronic signatures combined with appropriate safeguards may provide a feasible alternative. Beyond regulatory compliance, electronic consent improves version control [34], enhances comprehension through multimedia formats [35], reduces administrative workload and costs [36], and contributes to lowering the environmental footprint of clinical research [37,38].

4.3. Investing in robust and interoperable data infrastructures

Sustained investment in robust and interoperable data infrastructures is also essential. Fragmentation of hospital information systems and limited interoperability hinder the efficient use of real-world data. In France, the National Health Data System offers opportunities to complement cohort data with comprehensive medico-administrative information, supporting follow-up and outcome ascertainment for nested trials [12]. However, detailed clinical data must still be collected within cohorts.

Emerging artificial intelligence-based tools may assist with data extraction and preparation [39], but they require extensive human validation and should be viewed as complementary rather than substitutive solutions [40].

4.4. Innovating in research funding

The sustainability of cohort-based research infrastructures remains a major global challenge, largely driven by a structural mismatch between the continuous investment required to maintain high-quality platforms and the predominantly short-term, project-based nature of medical research funding [29]. Bridging this gap requires innovative funding models that move beyond transient support and provide long-term stability for cohort infrastructures.

In the United Kingdom, the National Institute for Health and Care Research supports institutional research capacity through the Research Capability Funding scheme, which allocates resources based on research activity and performance [41]. In the Netherlands, national funding schemes, particularly through ZonMw, have supported

cohort infrastructures and TwiCs-related initiatives, including platforms such as the Utrecht cohort for Multiple BREast cancer intervention studies and Long-term evaluation (UMBRELLA) [5]. The feasibility of implementing TwiCs beyond a single setting is further illustrated by oncology TwiCs conducted in the Netherlands [6,8] and by the ComPaRe e-cohort in France, where two TwiCs have been successfully funded [11].

In countries where dedicated infrastructure funding streams are not yet established, existing performance-based mechanisms could serve as an interim bridge. In France, eg, the SIGAPS (Système d'Interrogation, de Gestion et d'Analyse des Publications Scientifiques) and SIGREC (Système d'Information et de Gestion de la Recherche et des Essais Cliniques) systems could potentially be adapted by ring-fencing a modest proportion of clinical activity-related funding for cohort maintenance and platform coordination [42,43]. However, long-term sustainability will ultimately require dedicated national and international funding pathways for cohort infrastructures. Complementary strategies include targeted calls for nested trials that cover both trial-specific costs and proportional contributions to platform maintenance; public–private partnerships enabling governed industry access in exchange for infrastructure support, and fee-for-service models where research teams pay standardized fees per trial, with revenues reinvested into operations. Resource-pooling consortia could also distribute costs across institutions and enhance scalability.

At the international level, Horizon Europe [44] and the European Clinical Research Infrastructure Network [45] provide frameworks for cross-border collaboration and shared research infrastructures. Developing dedicated European funding streams for cohort-based platforms within these frameworks could strengthen the sustainability of TwiCs and enable larger, more diverse cohorts. Ultimately, securing the future of TwiCs requires a paradigm shift: cohorts should be recognized not as time-limited projects, but as strategic research infrastructures, analogous to biobanks or clinical trial networks, that warrant sustained investment. Coordinated advocacy, supported by evidence on scientific yield and operational efficiency, will be critical to fostering a robust international environment for clinical evidence generation.

5. Conclusion

The TwiCs design stands as a powerful evolution in clinical research, offering a pragmatic and efficient way to evaluate interventions with enhanced real-world applicability. Yet, research teams still face challenges in fully embracing this methodology.

Overcoming these challenges requires a multifaceted approach. Adapting the consent process to better align with ethical principles, potentially through electronic solutions, may facilitate broader acceptance. At the same

time, sustained investment in standardized data infrastructures and rigorous data management protocols is essential to unlock the potential of real-world data. Crucially, closing the financial gap will require both leveraging existing mechanisms as transitional support and advocating for dedicated funding streams tailored to the unique long-term needs of TwiCs. Progress will require coordinated efforts from researchers, funders, and policymakers to build a sustainable TwiCs ecosystem at both national and international levels.

CRediT authorship contribution statement

Claire Bahans: Writing – review & editing, Writing – original draft, Validation, Investigation, Conceptualization. **Claire Dossier:** Writing – review & editing, Investigation, Conceptualization. **Olivier Dunand:** Writing – review & editing, Investigation. **Cyrielle Parmentier:** Writing – review & editing, Investigation. **Bruno Ranchin:** Writing – review & editing, Investigation. **Gwenaelle Roussey:** Writing – review & editing, Investigation. **Stéphanie Tellier:** Writing – review & editing, Investigation. **Charlotte Samaille:** Writing – review & editing, Investigation. **Isabelle Vrillon:** Writing – review & editing, Investigation. **Céline Cluzan:** Writing – review & editing, Investigation. **Olivia Boyer:** Writing – review & editing, Investigation. **Vincent Guignonis:** Writing – review & editing, Writing – original draft, Validation, Investigation, Conceptualization.

Declaration of competing interest

None declared.

Data availability

No data were used for the research described in the article.

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